

Examples of Transcoder feature activations on the three-digit addition dataset

Notation. The label L_Y^x identifies transcoder feature Y at transformer layer x .

Models. All activations are extracted from Qwen2.5-3B-Instruct, a decoder-only transformer with 36 layers and a hidden dimension of 2048. Sparse autoencoders (SAE) trained on each MLP sublayer are drawn from the `mwhanna/qwen3-4b-transcoders` collection on HuggingFace.

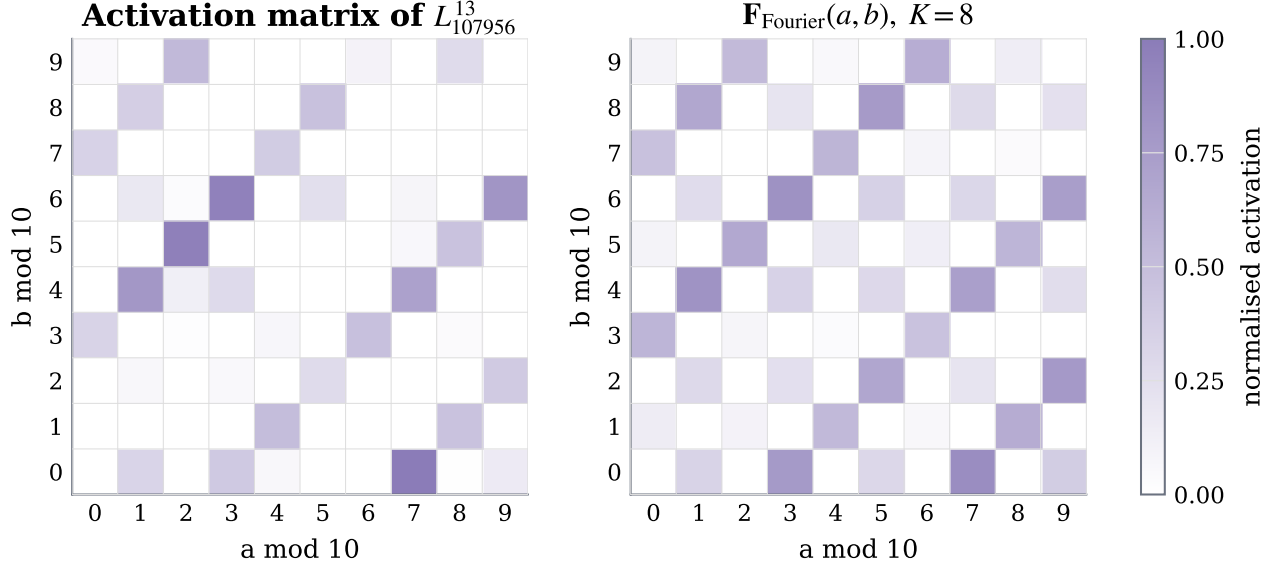
Dataset and prompt template. The addition dataset consists of 100 sampled prompts of the form "calc: a+b=", where $a, b \in \{100, \dots, 999\}$ are three-digit integers. Activations are recorded from the SAE at the position of unit digits of **b** operand. Each figure panel shows a 10×10 grid of mean feature activations indexed by $(a \bmod 10, b \bmod 10)$.

Fourier approximation. The activation grid is decomposed into real two-dimensional cosine modes on the digit torus via the discrete Fourier transform. The approximation $\mathbf{F}_{\text{Fourier}}(a, b)$ retains the top- K modes by amplitude, where each mode takes the form $A \cos(2\pi(ua + vb)/10 + \phi)$, with u, v integer frequencies along the a and b digit axes respectively. K is chosen as the smallest value that captures the dominant structure.

Equivalence. Where the $\equiv$ line appears below a Fourier formula, it gives the *theoretical* closed-form expression for the arithmetic concept the feature appears to encode. This expression is not fitted to data.

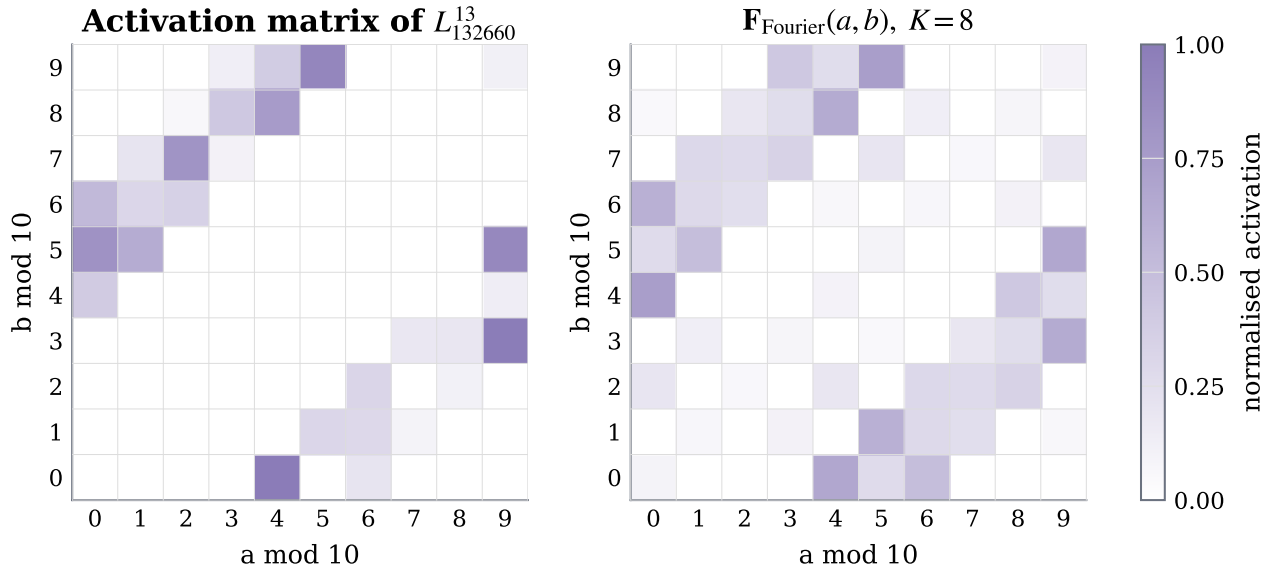
$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.13 + 0.24 \cos\left(2\pi \frac{5a + 5b}{10} - 3.14\right) + 0.16 \cos\left(2\pi \frac{2a - 2b}{10} - 3.14\right) + 0.16 \cos\left(2\pi \frac{3a - 3b}{10} - 0.02\right)$$

$$\equiv \mathbf{1}[|a - b| \equiv 3 \bmod 10]$$



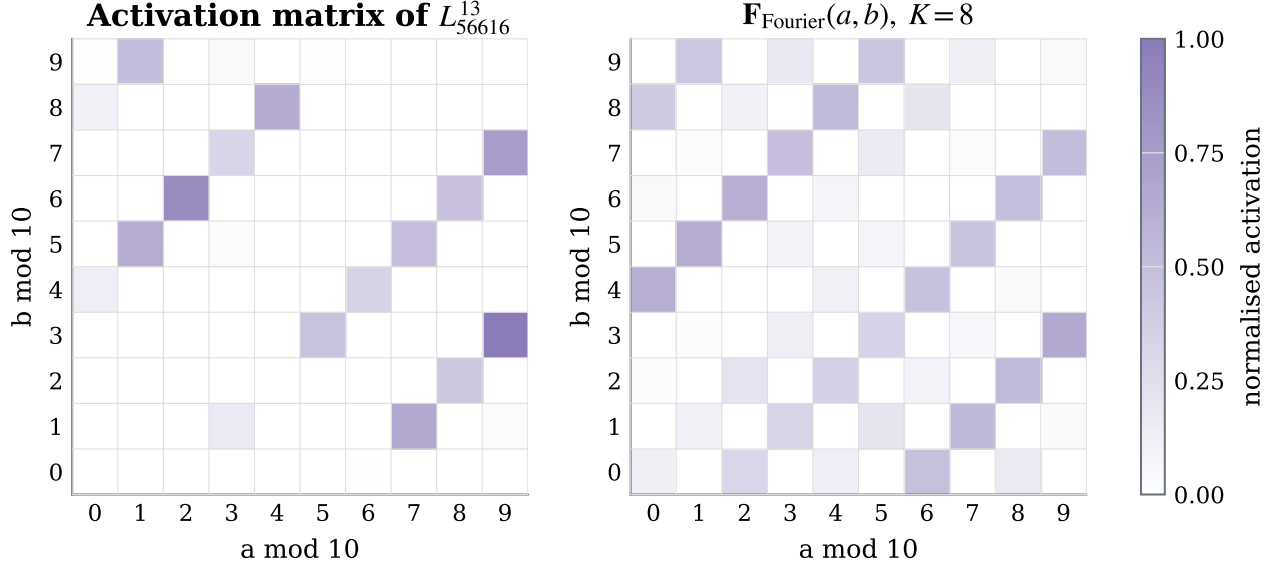
$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.12 + 0.2 \cos\left(2\pi \frac{a - b}{10} + 3.09\right) + 0.12 \cos\left(2\pi \frac{2a - 2b}{10} - 0.14\right) + 0.1 \cos\left(2\pi \frac{5a + 5b}{10}\right)$$

$$\equiv \mathbf{1}[a + b \bmod 10 \in \{4, 5, 6\}]$$



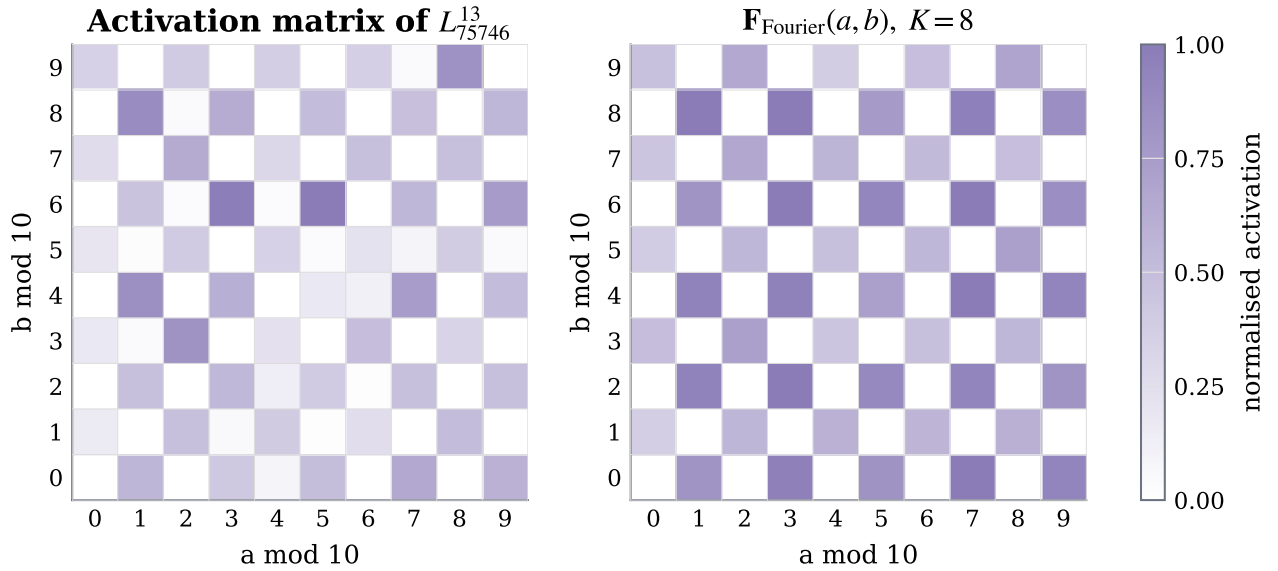
$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.08 + 0.16 \cos\left(2\pi \frac{5a + 5b}{10}\right) + 0.13 \cos\left(2\pi \frac{2a - 2b}{10} - 1.77\right) + 0.13 \cos\left(2\pi \frac{3a - 3b}{10} + 1.77\right)$$

$$\equiv \mathbf{1}[|a - b| \equiv 3 \bmod 10]$$



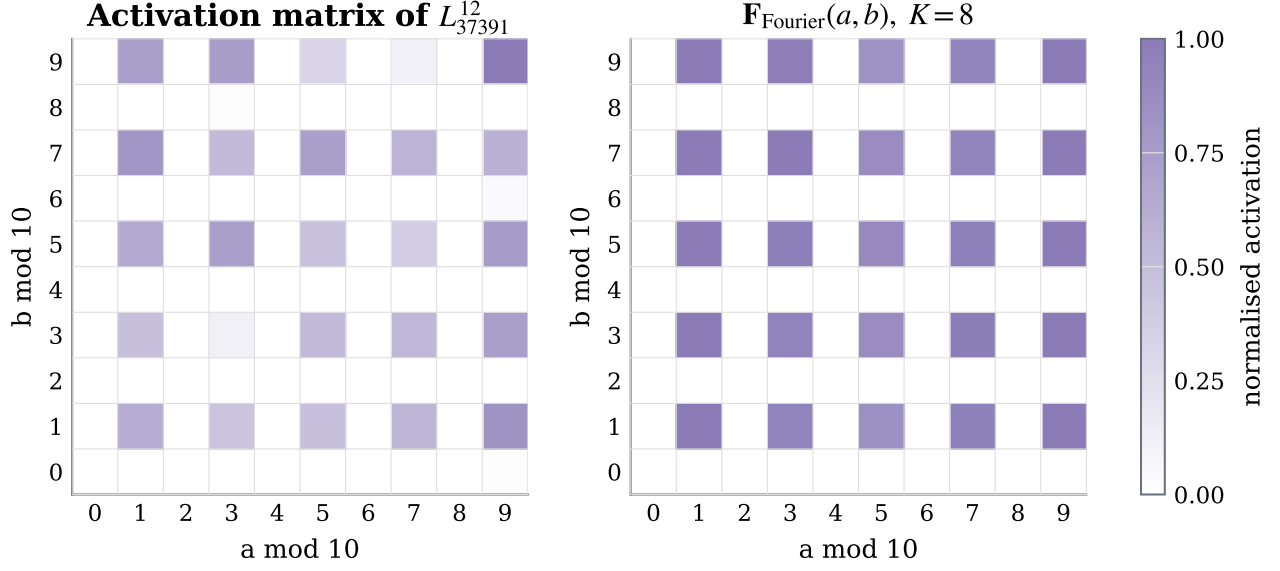
$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.25 + 0.48 \cos\left(2\pi \frac{5a + 5b}{10} + 3.14\right) + 0.1 \cos\left(2\pi \frac{5b}{10}\right) + 0.09 \cos\left(2\pi \frac{5a}{10} + 3.14\right)$$

$$\equiv (a + b) \bmod 2 = 1 \text{ for any } a, b$$



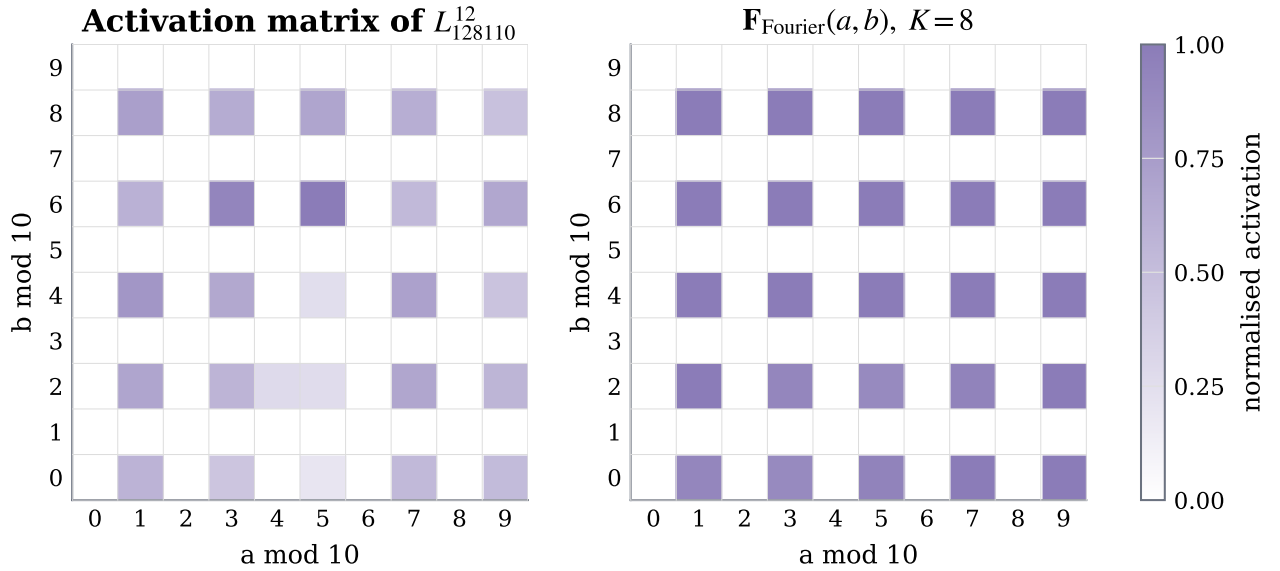
$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.14 + 0.29 \cos\left(2\pi \frac{5a}{10} + 3.14\right) + 0.29 \cos\left(2\pi \frac{5b}{10} + 3.14\right) + 0.29 \cos\left(2\pi \frac{5a + 5b}{10}\right)$$

$\equiv (a + b) \bmod 2 = 0$ for a, b odd



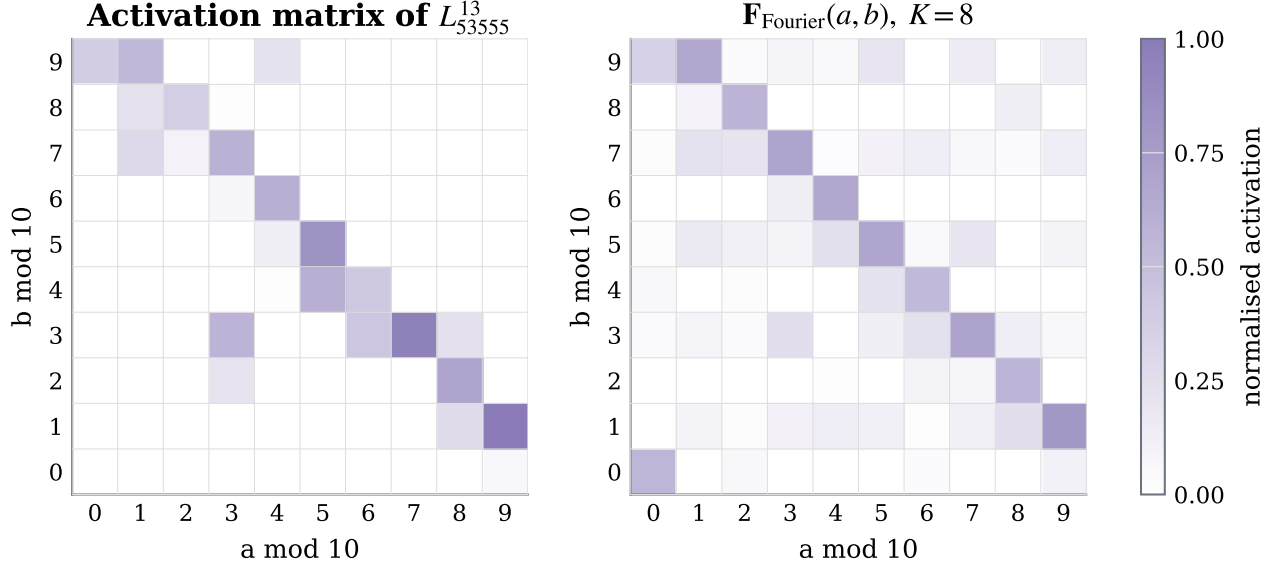
$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.15 + 0.3 \cos\left(2\pi \frac{5b}{10}\right) + 0.29 \cos\left(2\pi \frac{5a}{10} + 3.14\right) + 0.29 \cos\left(2\pi \frac{5a + 5b}{10} + 3.14\right)$$

$\equiv (a + b) \bmod 2 = 0$ for a odd, b even



$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.1 + 0.15 \cos\left(2\pi \frac{a+b}{10} + 0.21\right) + 0.14 \cos\left(2\pi \frac{2a+2b}{10} + 0.24\right) + 0.11 \cos\left(2\pi \frac{3a+3b}{10} + 0.44\right)$$

$$\equiv \mathbf{1}[a + b \equiv 0 \bmod 10] \text{ (carry overflow indicator)}$$



$$\mathbf{F}_{\text{Fourier}}(a, b) \approx 0.08 + 0.12 \cos\left(2\pi \frac{5a+5b}{10}\right) + 0.11 \cos\left(2\pi \frac{a+b}{10} + 0.15\right) + 0.11 \cos\left(2\pi \frac{3a+3b}{10} + 0.18\right)$$

$$\equiv \mathbf{1}[a + b \equiv 0 \bmod 10] \text{ (carry overflow indicator)}$$

